



# Vomiting in Infants and Children

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Vomiting is the uncomfortable, involuntary, forceful throwing up of food.

In infants, vomiting must be distinguished from spitting up. Infants often spit up small amounts while being fed or shortly afterward—typically while being burped. Spitting up may occur because infants feed rapidly, swallow air, or are overfed, but it may occur for no apparent reason. Vomiting is typically caused by a disorder. Experienced parents can usually tell the difference between spitting up and vomiting, but first-time parents may need to talk to a doctor or nurse.

Vomiting can cause significant loss of body fluid ([dehydration](#)). Sometimes children cannot drink enough to make up for lost fluid—either because they are continuing to vomit or because they do not want to drink. Children who are vomiting usually do not want to eat, but this lack of appetite rarely causes a problem.

(See also [Nausea and Vomiting](#) in adults.)

## Causes of Vomiting in Infants and Children

Vomiting can be beneficial by getting rid of toxic substances that have been swallowed. However, vomiting is most often caused by a disorder. Usually, the disorder is relatively harmless, but occasionally vomiting is a sign of a serious problem, such as a blockage in the stomach or intestine or increased pressure within the skull (intracranial hypertension).

### Common causes

Likely causes of vomiting depend on the child's age.

In **newborns and infants**, the most common causes of vomiting include

- [Gastroenteritis](#) (infection of the digestive tract) due to a virus
- [Gastroesophageal reflux](#)

In **older children**, the most common cause of vomiting is

- Gastroenteritis due to a virus

### Less common causes

In **newborns and infants**, some causes of vomiting, although less common, are important because they may be life threatening:

- Narrowing or blockage of the passage out of the stomach ([pyloric stenosis](#)) in infants aged 3 to 6 weeks
- A blockage of the intestine caused by birth defects, such as twisting (volvulus) or narrowing (stenosis) of the intestine
- Sliding of one segment of intestine into another ([intussusception](#)) in infants aged 3 to 36 months

[Food intolerance](#), [allergy to cow's milk protein](#), and certain uncommon [hereditary metabolic disorders](#) may also cause vomiting in newborns and infants.

In **older children and adolescents**, other causes of vomiting include serious infections (such as a [kidney infection](#) or [meningitis](#)), acute [appendicitis](#), a disorder that increases pressure within the skull (such as a [brain tumor](#) or a [serious head injury](#)), [gastroesophageal reflux disease](#) or [peptic ulcer disease](#), a slowly emptying stomach (gastroparesis), [food allergies](#), [eating disorders](#), [ingestion of a toxic substance](#) (such as large amounts of [acetaminophen](#), [iron](#), or alcohol), and [cyclic vomiting](#).

In adolescents, causes also include [pregnancy](#) and frequent use of cannabis (marijuana—[cannabinoid hyperemesis syndrome](#)).

## Evaluation of Vomiting in Infants and Children

For doctors, the first goal is to determine whether children are dehydrated and whether the vomiting is caused by a life-threatening disorder.

### Warning signs

The following symptoms and characteristics are cause for concern:

- Lethargy and listlessness

- In infants, inconsolability or irritability and bulging of the soft spots (fontanelles) between the skull bones
- In older children, a severe headache, stiff neck that makes lowering the chin to the chest difficult, sensitivity to light, and fever
- Abdominal pain, swelling, or both
- Persistent vomiting in infants who have not been growing or developing as expected
- Bloody stools
- Bright green or bloody vomit

### When to see a doctor

Children with warning signs should be immediately evaluated by a doctor, as should all newborns; children whose vomit is bloody, resembles coffee grounds, or is bright green; and children with a recent (within a week) head injury. If children appear uncomfortable even when not vomiting and their discomfort lasts more than a few hours, they should probably be evaluated by a doctor.

For other children, signs of dehydration, particularly decreased urination, and the amount they are drinking help determine how quickly they need to be seen. The urgency varies somewhat by age because infants and young children can become dehydrated more quickly than older children. Generally, infants and young children who have not urinated for more than 8 hours or who have been unwilling to drink for more than 8 hours should be seen by a doctor.

The doctor should be called if children have more than 6 to 8 episodes of vomiting, if the vomiting continues more than 24 to 48 hours, or if other symptoms (such as cough, fever, or rash) are present.

Children who have had only a few episodes of vomiting (with or without diarrhea), who are drinking at least some fluids, and who otherwise do not appear very ill rarely require a doctor's visit.

### What the doctor does

Doctors first ask questions about the child's symptoms and medical history. Doctors then do a physical examination. A description of the child's symptoms and a thorough examination usually enable doctors to identify the cause of vomiting (see table

[Some Causes and Features of Vomiting in Infants, Children, and Adolescents](#)

Doctors ask

- When the vomiting started

- How often it occurs
- What the vomit looks like (including its color)
- Whether it is forceful (projectile)
- How much is vomited

Determining whether there is a pattern—occurring at certain times of the day or after eating certain foods—can help doctors identify possible causes. Information about other symptoms (such as fever and abdominal pain), bowel movements (frequency and consistency), and urination can also help doctors identify a cause.

Doctors also ask about recent travel and injuries.

Doctors ask adolescent girls about missed periods, weight gain, and vomiting in the morning.

A physical examination is done to check for clues to possible causes. Doctors note whether children are growing and developing as expected.



## Some Causes and Features of Vomiting in Infants, Children, and Adolescents

| Cause<br>(listed from most<br>to least common) | Common Features*                                                                                                                                                                                                                                                                                                                                                     | Tests                                                                                                                                                              |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Gastroenteritis</u>                         | Usually with diarrhea<br>(which rarely is<br>bloody)                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                    |
|                                                | Sometimes a fever                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                    |
|                                                | Sometimes recent<br>contact with infected<br>people (as at a day<br>care center), with<br>animals at a petting<br>zoo (where <i>Escherichia</i><br><i>[E.] coli</i> may be<br>acquired), or with<br>reptiles (which may be<br>infected with<br><i>Salmonella</i> bacteria) or<br>recent consumption of<br>undercooked,<br>contaminated food or<br>contaminated water | A doctor's<br>examination<br><br>Sometimes<br>examination, culture,<br>or PCR testing of stool                                                                     |
| <u>Gastroesophageal<br/>reflux</u>             | Symptoms that occur<br>after feeding,<br>including fussiness,<br>spitting up, arching of<br>the back, crying, or a<br>combination                                                                                                                                                                                                                                    | A doctor's<br>examination                                                                                                                                          |
|                                                | Sometimes a cough<br>when lying down, poor<br>weight gain, or both                                                                                                                                                                                                                                                                                                   | Sometimes treatment<br>with medications to<br>suppress acid<br>production (if<br>symptoms are<br>relieved, the cause is<br>probably<br>gastroesophageal<br>reflux) |



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